

L01 LIGHT EXPOSURE | P

Intent : Provide indoor light exposure through daylight and electric light strategies.

Summary : This WELL feature requires projects to provide appropriate light exposure in indoor environments through lighting strategies.

Issue : Mammals function on an approximately 24-hour cycle, following what are referred to as circadian rhythms.¹ Light is the main driver of the circadian system, which controls body processes, such as digestion, the release of certain hormones, body temperature and sleep. Humans evolved their sleep schedules to be dependent on the natural day- night cycle. Electric light has only been widely used as a source of light for less than 200 years.² The increase in the use of electric lighting has led to indoor environments relying on electric light over daylight. However, reduced exposure to light has been linked to the onset of depression and impairment of cognitive function in individuals.^{3,4} Irregular sleep-wake cycles have been linked to poorer academic performance in college students.⁵ Studies have shown that light exposure during daytime has been linked to increases in productivity and sleep quality, as well as reduction in symptoms of depression in individuals.⁶⁻⁸

Solutions : Providing indoor access to adequate light can positively influence the productivity and mood of individuals, while supporting the alignment of their circadian rhythms with the natural day-night cycle.⁹⁻¹² Access to appropriate levels of light in indoor environments can be achieved through building design, façade design, space layout and lighting design. Windows, atriums and skylights are design features that can be utilized to increase daylight in a space. The interior layout of the space also has an impact on the daylight exposure received by users. For example, conference rooms can be added to the center of the floor plate, so that workstations can be situated near windows and have daylight exposure. Lighting strategies using electric lighting can be utilized to achieve required light exposure, when appropriate daylight exposure is not available.

PART 1 PROVIDE INDOOR LIGHT

For All Spaces except Dwelling Units & Guest Rooms:

Option 1: Daylight simulation

The project demonstrates, through computer simulations, that one of the following conditions are achieved:

- a. The total floor area of regularly occupied spaces achieves one of the following targets:

**Calculations per IES
LM-83¹³**

Calculations per Annex A of CEN 17037

$sDA_{150,50\%} > 30\%$

OR Target illuminance 200 lux is achieved for >30% of floor area throughout 50% of daylit hours of the year

- b. Common space that has unassigned seating for at least 15% of regular occupants at any given time achieves one of the following targets:

**Calculations
per IES LM-
83**

Calculations per Annex A of CEN 17037

$sDA_{300,50\%} > 75\%$

OR Target illuminance 300 lux is achieved for >30% of floor area and average illuminance 100 lux is achieved for >95% of floor area throughout 50% of daylit hours of the year

OR

Option 2: Interior layout

One of the following requirements is met:

- a. At least 30% of the regularly occupied area is within a 6 m horizontal distance of envelope glazing.
- b. Common spaces have unassigned seating and can accommodate at least 15% of regular occupants at any given time. At least 70% of all seating in the spaces is within a 5 m horizontal distance of envelope glazing.

OR

Option 3: Building design

One of the following requirements is met:

- a. The envelope glazing area is no less than 7% of the regularly occupied floor area.
- b.

The floor plate is no more than 20 m between opposite walls that each have envelope glazing, and there are no opaque obstructions higher than 1.22 m within a 6 m horizontal distance of the envelope glazing.

OR

Option 4: Circadian lighting design

The following requirement is met:

- a. The project meets the threshold of Tier 1 of Feature L03: Circadian Lighting Design.

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

For Dwelling Units & Guest Rooms:

Option 1: Daylight simulation

The project demonstrates, through computer simulations, that the following requirement is achieved:

- a. The total floor area of regularly occupied spaces of each dwelling unit achieves one of the following targets:

**Calculations per IES
LM-83¹³**

sDA_{150,50%} > 30%

Calculations per Annex A of CEN 17037

OR Target illuminance 200 lux is achieved for >30% of individual unit area throughout 50% of daylit hours of the year

OR

Option 2: Facade design

The following requirement is met:

- a. The envelope glazing area is no less than 7% of the regularly occupied floor area for each dwelling unit.

OR

Option 3: Circadian lighting design

The following requirement is met:

- a. The project achieves at least one point in Feature L03: Circadian Lighting Design.

REFERENCES

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